UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Class 011 · Lagrange's method of undetermined multipliers · compiled 15 October 2026

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1 Short notes

1 · Lagrange's method: the auxiliary function and the procedure
Optimise f subject to $\phi = 0$: $F = f + \lambda\phi$; solve $F_x = F_y = F_z = 0$ with $\phi = 0$ for x, y, z, λ. Advantages: no elimination; extends to many variables and constraints. Disadvantages: nature not decided; the equations are only necessary.

2 · Geometric problems: inscribed solids, nearest and farthest points
Box in sphere: $f = 8xyz \Rightarrow x = y = z = \frac{r}{\sqrt{3}}$, cube. Ellipsoid: $V_{\max} = \frac{8abc}{3\sqrt{3}}$. Distances: optimise the square. $x^2 - z^2 = 1$: nearest $(\pm 1, 0, 0)$. $5x^2 + 6xy + 5y^2 = 8$: distances 1 (min) and 2 (max). $(3, 4, 12)$ to unit sphere: 12 and 14.

3 · Applied problems: boxes, splitting a number, temperature
Open box, surface 432: $12 \times 12 \times 6$. $24 = 4 + 8 + 12$ for xy^2z^3. $400xyz^2$ on the unit sphere: 50. CW: $a^3x^2 + b^3y^2 + c^3z^2$ with $\sum \frac{1}{x} = 1$: $(a + b + c)^3$; $x^3 + 8y^3 + 64z^3$ with $xyz = 1$: 24.

2 Keyword index

12 terms; the number in brackets is the topic.

Advantage (1). No elimination; many variables and constraints.

Auxiliary function (1). $F = f + \lambda\phi$.

Constrained extremum (1). Extremum of f on $\phi = 0$.

Continued product (3). xy^2z^3 for first, square of second, cube of third.

Disadvantage (1). Nature not determined; conditions only necessary.

Four equations (1). $F_x = F_y = F_z = 0$, $\phi = 0$.

Hyperbolic cylinder (2). $x^2 - z^2 = 1$; nearest points $(\pm 1, 0, 0)$.

Inscribed box (2). $V = 8xyz$; cube in a sphere.

Lagrange multiplier λ (1). Undetermined constant multiplying the constraint.

Open box (3). Surface $xy + 2yz + 2zx$.

Parallel gradients (1). $\nabla f = -\lambda \nabla \phi$ at the optimum.

Squared distance (2). Optimise $x^2 + y^2 + z^2$ instead of the distance.

3 Rapid recall

1	Auxiliary function?	$F = f + \lambda\phi$.
2	Equations to solve?	$F_x = F_y = F_z = 0$ and $\phi = 0$.
3	Main advantage?	No need to eliminate a variable; extends to more variables and constraints.
4	Main disadvantage?	Does not tell maximum or minimum.
5	Largest box in a sphere?	A cube, half-edge $\frac{r}{\sqrt{3}}$, $V = \frac{8r^3}{3\sqrt{3}}$.
6	Largest box in an ellipsoid?	$V = \frac{8abc}{3\sqrt{3}}$.
7	Nearest points of $x^2 - z^2 = 1$ to the origin?	$(\pm 1, 0, 0)$.
8	Distances from the origin to $5x^2 + 6xy + 5y^2 = 8$?	Max 2, min 1.
9	Distances of $(3, 4, 12)$ from the unit sphere?	Max 14, min 12.
10	Open box, surface 432 cm²?	$12 \times 12 \times 6$ cm.
11	Divide 24 for max xy^2z^3?	4, 8, 12.
12	Highest $400xyz^2$ on the unit sphere?	50.
13	Extreme of $x^3 + 8y^3 + 64z^3$, $xyz = 1$?	24 at $(2, 1, \frac{1}{2})$.

Mnemonic (study aid). *"Add, differentiate, divide out* λ*, constrain."* $F = f + \lambda\phi$; $F_x = F_y = F_z = 0$; equate the expressions for λ; finish with $\phi = 0$.

4 Last 10 minutes before the exam

Top facts

1. $F = f + \lambda\phi$; solve $F_x = F_y = F_z = 0$, $\phi = 0$.
2. Make λ the subject (or multiply by x, y, z) to get relations like $x = y = z$.
3. Nature comes from the problem, not from the method.
4. Box in sphere ⇒ cube; in ellipsoid ⇒ $\frac{8abc}{3\sqrt{3}}$.
5. Distances: optimise the square. $5x^2 + 6xy + 5y^2 = 8$: 1 and 2.
6. $400xyz^2$ on the unit sphere: 50. Open box 432 cm²: $12 \times 12 \times 6$. 24 ⇒ 4, 8, 12.

Top formulas

E1.1 · Auxiliary (Lagrange) function ESA ★★★

$$F(x, y, z) = f(x, y, z) + \lambda\,\phi(x, y, z)$$

GIVES One function whose stationary points are the constrained stationary points.

APPLIES Optimise f subject to $\phi(x, y, z) = 0$.

NOTE Write the constraint with everything on one side: $x^2 + y^2 + z^2 - r^2$, $xy + 2yz + 2zx - 432$.

S011·3 · T011a 00:01:08

E1.2 · The four equations ESA ★★★

$$F_x = f_x + \lambda\phi_x = 0, \quad F_y = f_y + \lambda\phi_y = 0, \quad F_z = f_z + \lambda\phi_z = 0, \quad \phi = 0$$

GIVES x, y, z and λ (four unknowns, four equations).

APPLIES Always; with two variables drop F_z (three equations).

NOTE Equivalently $\nabla f = -\lambda\nabla\phi$: the gradients are parallel (added for clarity).

S011·3 · T011a 00:02:17 · T011a 00:02:30

E1.3 · Solving trick: make λ the subject ESA ★★

$$\lambda = -\frac{f_x}{\phi_x} = -\frac{f_y}{\phi_y} = -\frac{f_z}{\phi_z}$$

GIVES Relations between x, y, z without λ.

APPLIES When $\phi_x, \phi_y, \phi_z \neq 0$.

NOTE Or multiply the equations by x, y, z and subtract (slides 011·6, 011·11).

S011·13 · S011·14 · T011b 00:07:24

E2.1 · Largest box in a sphere and in an ellipsoid ESA ★★★

$$x^2+y^2+z^2=r^2:\ x=y=z=\tfrac{r}{\sqrt{3}},\ V_{\max}=\tfrac{8r^3}{3\sqrt{3}}$$
$$\tfrac{x^2}{a^2}+\tfrac{y^2}{b^2}+\tfrac{z^2}{c^2}=1:\ x=\tfrac{a}{\sqrt{3}},\ y=\tfrac{b}{\sqrt{3}},\ z=\tfrac{c}{\sqrt{3}},\ V_{\max}=\tfrac{8abc}{3\sqrt{3}}$$

GIVES Half-edges and the largest volume $V=8xyz$.

APPLIES Box centred at the origin, edges $2x, 2y, 2z$ parallel to the axes.

NOTE The sphere is the ellipsoid with $a=b=c=r$: the box is a cube.

S011·5 · ESA May-24 1c · ESA Jan-24 s3 1a · ESA Jan-24 s4 1a

E2.2 · Distance problems: use the square ESA ★★★

$$f=x^2+y^2+z^2\ \ (\text{origin}),\qquad f=(x-x_0)^2+(y-y_0)^2+(z-z_0)^2$$

GIVES The squared distance, which has its extrema at the same points as the distance.

APPLIES Nearest or farthest point on a curve or surface.

NOTE Take the square root at the end.

ADDED FOR CLARITY S011·8 · ESA Jul-23 1c · CW U1 P36